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22 January 1982
SUPERSEDING
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MILITARY SPECIFICATION

CANNON 8 INCH HOWITZER: M201/M201A1

This specification is approved for use by to US army amament Reseach and Development Command, Department of the Army, and is available for use by all Department and Agencies of the Department of Defense.

1. sqope

1.1 This specification covers Cannon, 8 Inch Howitzer: M201 and m201a1 for use with Howitzers, Heavy, Self-Propelled: 8 Inch, m110a1 and Ml10A2. The M201 cannon consists of a tube assembly, breech mechanism assembly, and firing mechanism assembly. The M201Al has, in addition, a muzzle brake, and a thrust collar.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otheRWise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN : DRDAR-LCB-SAS, Watervliet, N.Y. 12189 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Military

MIL-C-13931	Cannons, General Specification for
MIL-P-14232	Parts, Equipment and Tools for Amy Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD- 109	Quality Assurance Terms and Definitions
MIL-STD-721	Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety.
MIL-STD-882	System Safety Program Requirements
MIL-STD-1235 -	Single and Multi-level continuous Sampling Procedures and Table for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications from part of this specification to the extent specified herein.

DRAWINGS

U S ARMY Armament Research **and** Development Command

EPL 11578719 -	Engineering Parts List, Cannon 8 Inch Howitzer:	M201
EPL 11579521-	Engneering Parts List, Cannon 8 Inch Howitzer:	M201A1
PDS 11578719-	Packaging Data Sheet, Cannon, 8 Inch Howitzer:	M201
PDS 11579521-	Packaging Data Sheet, Cannon, 8 Inch Howitzer:	**M201A1**

U.S. Army Armament Materiel Readiness Command

DL 11578719 - List of Inspection Equipment for Cannon, 8 Inch Howitzer: M201
DL 11579521 - List of Inspection Equipment for Cannon, 8 Inch Howitzer: M201a1
Master LQAP 11578719 - Master List of Quality Assurance Provisions for Cannon, 8 Inch Howitzer: M201
Master LQAP 11579521 - Master List of Quality Assurance Provisions for Cannon, **8** Inch Howitzer: M201Al
QAP-APPENDIX-WV - General Quality Assurance Provisions

PUBLICATION

DA Foxm 2408-4 - Weapon Record Data

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer),

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence; however, applicable engineering drawings shall take precedence.

3. REQUIREMENTS

3.1 First aRTIcle. When specified, a sample sHAll be subjected to first article inspection (see 4, and 6).

3.2 General requirements. Cannon and parts thereof shall conform to the drawings listed in Engineering Parts List 11578719 for M201, Engineering Parts List 11579521 for M201A1, the requirements of this specification and MIL-C-13931.

3.2.1 Definitions.

3.2.1.1 - RAM-D terms and definitions. Reliability, availability, maintainability and durability terms and definitions used in this specification are in accordance with MIL-STD-721. Durability is further defined in 3.2.1.1.1.

3.2.1.1.1 Durability. Durability is the probability that an item will successfully survive its projected life, overhaul point, or rebuild point (Whichever is the more appropriate durability measure for the item) without a durability failure. A durability failure is considered to be malfunction that precludes further operation of the item and is great enough in cost, safety, or time to restore, what the item must be replaced or rebuilt.

3.2.1.2 EFC (Equivalent Full Charge). An EFC round is a Zone 9M188Al charge, fired at 70°F, utilizing an M106 projectile.

3.2.1.3 Failure. For the purpose of accessing reliability, a failure is defined as any malfunction which the operator or crew cannot or is not authorized to remedy by adjustments, repair, or replacement action within five minutes, using the controls, Basic Issue Items (BII), Items troop Installed or Authorized (ITIA) and parts authorized to the crew, which causes or would cause: 1. inability to commence the cessation of any rode-of operation. 2. a critical or catastrophic hazard to personnel or equipment as defined by MIL-STD-882. For related malfunctions only the primary malfunction will be counted against reliability (see APPENDIX 30.1.1).

3.2$1.4 UpPer pressURE limit. The upper pressure limit (see 3.4.1) is that value of the maximum pressure which is specified in the propellant specification as the upper limit of average pressure which may be developed by an acceptable propellant in the form of propelling charges which will impart the specified muzzle velocity to the specified projectile at its standard zone weight.

3,2.2 Reliability, availability, maintainability, durability (RAM-D). Since availability and maintainability requirements are specified for the M110A2 Howitzer system only, these attributes cannot be apportioned for the cannon; however, the reliability and durability requirements are specified for the cannon.

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3.2.2.1 Reliability. The cannon, as a component of a fully operational 8 Inch Howitzer: M110A1/A2, shall have for the Operational Mode Summary - Mission Profile (OMS-MP) a reliability of 1600 Equivalent Full Charge (EFC) Mean Rounds Between Failure (MRBF) with 80 percent confidence (see 4.8.3.4). Failures are defined in 3.2.1.3. The scoring criteria for assessing the reliability requirements are stated int the APPENDIX.

3.2.2.1.1 Mission profile. A typical 48 hour mission consists of firing mixed charge rounds equivalent to 94EFC rounds.

3.2.2.1.2 Climatic. The cannon will be employed in all types of areas, terrain and climates in which the supperted units may operate. This requires o peration in geographical areas which include temperature extremes from 125°F to -50°F .

3.2.2.2 Durability. The cannon parts and components shall have the following durability:

a. Breech ring assembly - 10,000 EFC rounds,
b. Breechblock assembly - 10,000 EFC rounds.
c. Tube assembly - 10,000 EFC rounds or 0.135 inch bore wear measured at 46.0 inches from the rest face of the tube.
d. Muzzle brake - equal to tube life.
e. All other parts and components will be replaced as required in scheduled maintenance.

3.2.3 Protective finishes. When required, protective finishes shall be in accordance with the applicable drawings.

3.3 Performance characteristics. The cannon **shall** perform the firing, pressure sustaining, and projectile launching functions of the Howitzer, Heavy, Self-Propelled; 8 Inch, M110a1 and M110A2. The breech mechanism, firing mechanism and other moving parts shall function smoothly without interference, erratic movement or malfunction.

3.3.1 Breech mechanism. When the breech is open, a pull on the operating lever to the left, assisted by the counterbalance after the hinge pin arm passes dead center, shall-swing the breechblock into the breech ring. This action allows the roller on the breechblock to strike the cam and rotate the breechblock clockwise to engage the sectored threads. An upward push on the operating-handle shall complete the rotation of the breechblock and lock the breech. The operating-handle shall latch in the closed position. To open the breech a rearward pull on the operating lever, after being unlatched, shall rotate the breech block counterclockwise to disengage the sectored threads. A pull on the lever to the right, assisted by the counterbalance after the hinge-pin passes dead center, shall swing the breechblock out of the breech ring. The counterbalance shall retain the breech mechanism in the open position.

3.3.2 Firing mechanism block assembly When the breechblock is rewed to close the breech, the lowered block assembly sha11 be automatically cammed from the extract

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position to the load position, and the extractor shall be cammed forward against the rear of the spindle. The block assembly shall be capable of moving to the closed position only when the breechblock is closed. In this position the block assembly shall also be capable of being moved manually into the load and extract positions. When the breechblock is mved to the open position, the block assembly shall slide in the slot in the housing assembly into the extract position. As the extract position is reached, the extractor shall forcibly eject the primer out of the spindle chamber.

3.3.3 Obt'mater. The obturator surfaces and the bearing of obturator parts shall comform to the requirements prescribed in MIL-C-13931.

3.3.4 Firing mechanism. When assembled in the cannon the firing mechanism assembly shall fire the primer as its lever is pulled by the attached lanyard. When th lanyard is released, the mechanism shall position itself for the next cycle.

3.3.5 Primer headspace Primer headspace, the distance between the front face of the block assembly and the rear face of the obturator spindle, shall- be not less than 0.052 inch nor more than 0.056 inch.

3.3.6 Firing pin protrusion and retraction.- The firing pin protrusion measured from the front face of the block shall be no less than 0.114 inch and not more than 0.118 inch in the fired position. Firing pin retraction behind the front face of the block shall. be not less than 0.012 inch.

3.3.7 Functional location of recoil cylinder rod hole in the breech ring to rails.

relation to the rails on the assembled cannon, shall be functionally checked-with inspection equipment designed from dimensions, either explicit, derived, or calculated in conformance with the applicable drawings listed in EPL 11578719 or EPL 11579521. The functional check sha1l show that the cannon will assemble to the mount without interference.

3.4 Performance.

3.4.1 High pressure resistance. The cannon shall be capable of withstanding a chamber pressure of 44800 + 1600 psi at 70F when tested as specified in 4.7.6, 4.8.3.1, and 4.8.3.2. The-pressure range listed represents 113 ± 4%of the upper pressure limit (see 3.2.1.4).

3.4.2 Material soundness. After testing, all parts shall be free from cracks and fractures and shall comply with soundness requirements specified on the applicable drawings.

3.4.3 Bore enlargement. Bore diameters measured across lands, after firing no more than 8 rounds, shall not exceed the basic (minimum) diameter specified on the applicable drawing by more than the amounts in Table I.

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TABLE 1. Bore enlargement.

Distance forward of commencement of rifling	0.10 inch	1. inch	1 cal	2 cals to muzzle
Amount (in inches)	**0.009**	**0.009**	0.008	**0.006**

NOTE : Interpolate to determine the allowable enlargement between the distances listed.

4. QUality ASSUrance PROVISiONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspec-cion requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use its own or any other facilities suitable for the performance of the inspection requirments specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The inspections set forth in the specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of its responsibility for assuring that all supplies submitted to the Government for acceptance conform to all requirements of the contract.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in the specification are in accordance with MIL-STD-109.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

First article inspection (see 4.4).
a. First article initial production test FA-IPT (see 4.5).
c. Reevaluation of critical components (see 4.6).
d. Quality conformance inspection (see 4. 7).

4.4 First article inspection. Unless otherwise specified in the contract , first article cannon and specified parts thereof shall be submitted for inspection in accordance with QAP-APPENDIX-WVA and applicable item QAPs. First article cannon and parts thereof shall be representative of the processes to be used during quantity production. QAPs are listed in Master LQAP 11578719 or Master LQAP 11579521, as applicable. First article quantities shall be as specified in the applicable item QAPs.

4.5 First article initial production test (FA-IPT). When specified in the contract (see 6.1.1) first article cannon shall be subjected to first article initial production testing (FA-IPT), by a Government activity to assure compliancetith the requirements of this specification and the contract.

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4,5.1 Sample. Four first article cannon, two additional breech ring assemblies, and two additional breechlock assemblies shall be randomly selected from the first month's production. First article cannon and parts thereof shall be representative of processes to be used during quantity production.

4.5.2 Inspection provision. First article cannon and parts thereof shall be subjected to 100 percent inspection of all dimensional and physical characteristics, quality conformance inspection (see 4.7), including proof acceptance (see 4.7.6), and any additional requirments listed in the contract, These cannon shall have successfully passed proof acceptance (see 4,7,6) through proof firing prior to commencement of FA-IPT. All firing tests shall be conducted by the Government or any agency specified nment at a Government specified test site. The results of the quality conformance inspection shall be used as the initial production records of the first article sample.

4.5.3 **Tests.** Unless otherwise specified herein or in the contract, first article cannon tests listed in this specification shall be performed in a fully operational HOWITZER, HEAVY, SELF-PROPELLED: 8 INCH, M11OA1 *M110A2* **and** shall not normally be integrated with other tests. Integration with Operational Testing by the user is allowable provided all components of the system conform to technical requirements. First article initial production tests shall include the following:

4.5.3.1 Climatic test. The climatic test shall be conducted on four cannon using the test method of 4.8,3.3. If the requirements of 3.2.2.1.2 are not complied with, acceptance shall be deferred and 4.9 shall apply.

4.5.3. 1.1 Pre-reliability test inspections. Upon completion of climatic tests, all examinations (see 4.7.4), and functional "inspections (see 4.7.5) shall be repeated before reliability tests (see 4.5.3.2) and durability tests (see 4.5.3.3) are performed.

4.5.3.2 Reliability test. The reliability test shall be conducted on two cannon using the test method of 4.8. 3.4. If the requirements of 3.2.2.1 are not complied with: acceptance shall be deferred and 4.9 shall apply.

4,5.3.3 Durability test. The durability test shall be conducted on two cannon using the test method of 4.8.3.5. If the requirements of 3.2.2.2 are not complied with, acceptance shall be deferred and 4.9 and 4.10 shall apply as applicable.

4.6 Reevaluation inspection of critical components. Reevaluation inspection may be performed on critical components during the course of the contract when: (a) a new supplier is utilized; (b) a time lapse has occurred since the item was last furnished by a supplier; (c) changes are made in the manufacturing process or material of the item; or (d) a design change is rode, and when the potential impact of these factors is judged significant enough by the responsible product assurance element of the procuring activity to Warrant testing.

4.6.1 Critical components. The following items of the canner are classified as

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critical components and are subject to reevaluation inspection: tube, breech ring, breech bushing, breechblock, and Muzzle brake (m201A1 cannon only).

4.6.2 Reevaluation inspection. Reevaluation inspection shall be as specified in Table II. ExamiNAtions and functional inspections shall be performed before and after completion of specified tests.

4.7 quality conformance inspection.

4.7.1 Inspection provisions. Cannon and parts thereof shall be inspected for conformance to the requirements of this specification, MIL-C-13931, QAPs listed in Master LQAP 11578719 or Master LQAP 11579521, as applicable, and QAP-.WPEVIIX-WVA.

4.7.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-c-13931.

4.7.3 Inspection equipment. Data lists containing drawing numbers of inspection equipment used in the inspection of cannon and parts thereof are referenced in List of Inspection Equipment DL 11578719 or DL 11579521, as applicable.

4.7.3.1 Acquisition, maintenance, and disposition. Unless otherwise specified (see 6.1.1), responsibility for acquisition, maintenance, and disposition of inspection equipment referenced in 4.7.3 and for all other inspection equipment referenced in applicable specifications shall be in accordance with MIL-I-45607.

4.7.3.2 Accuracy of standard measuring equipment. When commercial or modified commercial inspection equipment is used, it shall be used measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.7.3.2.1 Accuracy of chamber pressure measuring devices. The devices for measuring chamber pressures (see 4.8.3. lc) shall have the following accuracy:

Device	Maximum error of measurement
Copper crusher gages	t 3% of reading
pressure transducers	± 2% of reading

4.7.4 Quality conformance examinations.

4.7.4.1 Primer headspace. The primer headspace of each cannon shall be measured using the examination method of 4.8.1.1. Failure to comply with the requirements of 3.3.5 shall be cause for rejection.

4.7.4.2 Firing pin protrusion and retraction. The firing Din protrusion and retraction of each cannon shall be measured using the examination method of 4.8.1 2. Failure to comply with the requirements of 3.3.6 shall be cause for rejection.

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TABLE II. -- Reevaluation inspections.

Inspections	Requirement Paragraph	Examination or Test Method Paragraph	1	Criti
Examinations				
Primer headspace	3.3.5	4.8.1.1		x
Firing pin protrusion and retraction	3.3.6	4.8. 1.2		X
Borescope	3.4.2	4.8.1 .3.1	X	
Magnetic particle	3.4.2	4.8.1.3.2	X	X
Function 1 inspection				
Breech mechanism and firing mechanism	3.3.1 3.3.2 3.3.4	4.8.2.1		x
Tests				
Proof acceptance	3.4.1 3.4.2 3.4.3	4.8.3.1 4.8.3.2	X	
Climatic	3.2.2.1.2	4.8.3.3	x	X
Durability	3.2,2.2	4.8.3.5		

* To be determined by the product assurance organization of the responsible proc
on a case by case basis.

**(M201Al cannon only)

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4.7.4.3 Material soundness.

4.7.4.3.1 Borescope. Each cannon tube bore shall be examined with a borescope, using the examination method of 4.8.1,3.1. Failure to comply with requirements of 3.4.2 shall be cause for rejection.

4.7.4.3.2 magnetic particle. Each cannon tube, tube assembly (11578720), breech ring, breech bushing, breechblock, and muzzle brake (M201a11 cannon only) shall be magnetic particle inspected using the examination method of 4.8.1.3.2. Failure to comply with the requiremmts of 3.4.2 shall be cause for rejection.

4.7.4.4 Packaging inspection. Unless otherwise specified, inspection to determine compliance with packaging requirements of the applicable packaging data sheet for the level designated in the contract shall be as specified in MIL-P-14232.

4.7.4.5 Workmanship. Cannon and parts thereof shall be examined for workmanship in accordance with MIL-C-13931.

4.7.5 Quality conformance functional inspections.

4.7.5.1 Breech mechanism and firing mechanism. Each breech mechanism, firing mechanism block assembly and firing mechanism shall be inspected using method of 4.8.2.1. Failure to comply with the requirements of 3,3.1, 3,3.2, and 3.3.4 shall be cause for rejection.

4.7.5.2 Obturator. The obturator surfaces and the bearing of obturator parts of each cannon shall be inspected using the method of 4.8.2.2 and MIL-C-13931. Failure to comply with the requirements of 3.3.3 and MIL-C-13931 shall be cause for rejection.

4.7,5.3 Functional location of rod hole in breech ring to cannon rails. Each assembled cannon shall be inspected for functional location of the recoil-cylinder-rod-hole in the breech ring relative co the rails of the cannon assembly, using the method of 4.8.2.3. Failure to comply with the requirements of 3.3.7 shall be cause for rejection.

4.7.6 Proof acceptance. Cannon shall be proof accepted though proof firing, simulation tests or both. Cannon to be proof fired shall be tested using the method of 4.8.3.1. Failure to comply with the requirements of 3.4 shall be cause-for rejection. Cannon breech mechanisms to be simulation proof tested shall be tested using the method of 4.8.3.2. " Failure to comply with the requirenmts of 3.4,1 and 3.4.2 shall be cause for rejection. Proof firing shall be done on a sampling basis and shall be conducted as follows:

a. Using sampling plan CSP-l of MIL-STD-1235 and an AQL of 1.0, the Product Assurance Directorate at Watervliet Arsenal shall designate the appropriate initial consecutive quantity (i) and the sampling frequency (f) (see 6.1.1) in the contract.

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b. The first consecutive "i" quantity of cannon produced in the particular buy slhall be proof fired. When the consecutive (i) quantity has been satisfied, sampling shall be put into effect using the "f" sampling frequency. All cannons not proof fired shall have the breech mechanism similation proof tested in the Product Assurance Weaopon Test Facility at Watervliet Arsenal or other Government - approved facility.

c. Successive production buys shall not require "i" consecutive quantities; however, if the time lag between two successive buys exceeds one month, or if a new cannon or tube contractor is utilized, a new "i" consecutive quantity shall be proof fired.

At the discretion of the Product Assurance Directorate at Watervliet Arsenal, additional "i" consecutive quantities may be required due to any interruption of the normal production flow caused by a labor, manufacturing process equipment, or material problem, or by an adverse natural phenomenon. .

Proof fired cannon, and simulation proof tested breech mechanisms **shall** be furnished with data on DA Form 2408-4 (Weapon Record Data) as specified by SARWV-QAR.

4.8 Methods of inspection.

4.8.1 Examination methods.

4.8.1.1 Primer headspace **The primer headspace shall** be gaged on each cannon with the applicable gage referenced in List of Inspection Equipment, DL l1578719 0r DL 11579521, as applicable, or Government - approved equivalent gage.

4.8.1.2 Firing pin protrusion and retraction. Firing pin protrusion and retraction shall be gaged on each cannon with the applicable gages referenced in List of Inspection Equipment, DL 11578719 or DL 11579521, as applicable, or Government - approved equivalent gages.

4.8.1.3 Material soundness.

4.8. 1.3.1 BorescoPe. The inner surface of each tube or tube assembly (11578720) shall be examined a white light borescope, and when specified in applicable QAPs, by a black light borescope in accordance with applicable drawings and specification.

4.8. i.3.2 magnetic particle. The following shall be magnetic particle inspected in accordance with applicable drawings and specifications:

a. The outer surface of each tube or tube assembly (1157872 0).

b. The inner surface of each tube when a black light borescope is used (see 4.8.1.3.1).

c. All surfaces of each breech ring, breech bushing, breechblock and muzzle brake (m201Al cannon only).

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4.8.2 Functional inspection methods.

4.8.2.1 Breech mechanism and firing mechanism. Functional inspection shall be performed either manually or through the use of a functional tester. A minimum of five complete cycles of breech operation shall be performed on each breech mechanism and firing mechanism:

a. Each time the breechblock assembly reaches the fully closed position, the firing mechanism block assembly shall be cycled between the load, the extract, and the firing positions.

b. Each time the firing mechanism block assembly is cycled into the firing position, the firing mechanism lever shall be pulled rearward and released; pulled clockwise and released; and pulled counterclockwise and released.

c. After each complete cycle of breech operation, the breech mechanism operating handle shall be observed to verify that it latches in the closed position.

4.8.2 2 Obturator. The obturator surfaces of each cannon shall be examined for cleanliness. The bearing of the obturator parts of each cannon shall be inspected in accordance with MIL-C-13931.

4.8.2.3 Functional location of rod hole in breech ring to cannon rails. Each assembled cannon shall be inspected for functional location of the recoil-cylinder-rod-hole in the breech ring relative to the rails of the cannon assembly, using the applicable gage referenced in List of Inspection Equipment, DL l1578719 or DL 11579521, as applicable; or Government - approved equivalent gage.

4.8.3 Test methods.

4.8.3.1 Cannon proof test. Proof firing of cannon shall be conducted at a Government specified test site in accordance with the following procedures:

a. Cannon shall be proof fired once at 44,800+ 1600 psi,

b. A facility mount, breech, muzzle brake may be used, if necessary.

c. Chamber pressure shall remeasured (see 4.7.3.2.1) and recorded.

d. Cannon shall be inspected for proper functioning (see 4.8.2.1), and obturator (see 4.6.2.2).

e. Bore enlargement (see 3.4.3 and Table I) shall be measured with a star gage, an air gage, an electronic gage, or other Government - approved gage. Results shall be recorded.

f. Cannon shall be borescoped (see 4.8.1.3.1), and magnetic particle inspected (see 4.8.1.3.2).

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g. Cannon found to be defective at the proving ground, shall be reported to the Product Assurance Directorate, Watervliet Arsenal, Watervliet, NY 12189. The test agency shall forward an Equipment Performance Report (EPR), incident to the defect within 12 hours for critical incident is considered an incident of concern which involves a catastrophic or critical hazard affecting the safety of personnel, performance of man-machine interface or makes test suspension advisable.

4.8.3.2 Breech mechanism simlation proof test. Breech mecanisMs of cannon not proof fired shall be simulation proof tested in the Product Assurance Weapon Test Facility at Watervliet Arsenal or other Government - approved facility, in accordance with the following procedure:

a. Breech mechanisms shall be tested by suitable hydrostatic pressure test equipment at a proof pressure of 44,800 ± 1600 psi.

b. Cycle time to achieve proof pressure and release shall not exceed five minutes. Proof pressure shall be imnediately released upon its attainment; however, a dwell time at proof pressure of 0.4 sec. is permissible.

(1) When difficulty is encountered in attaining proof pressure, the number of attempts in which 25 percent or less of the proof pressure is attained, **shall not be counted. The number** Of attempts in which over 25 percent of the proof pressure is attained, shall not exceed five.

If further investigation does not reveal **why** proof **pressure** could not be attained, the breech mechanism shall be subjected to proof firing as part of the cannon (see **4.8.3.1).**

(3) The peak pressure shall be recorded for each test and attached to inspection records identifiable with the specific breech mechanism undergoing test.

c. After proof pressure testing, the breech mechanism shall be examined for looseness or interference of fit and other defects of workmanship or function.

d. The breech mechanism shall be disassembled and examined for material soundness. Each component or subassembly shall be visually examined to assure freedom from cracks and fractures. All surfaces of each breechblock. breech bushing. and breech ring *shall be magnetic* particle inspected in accordance with applicable drawings and specifications.

4.8.3.3 Climatic test. The climatic test shall be conducted on four cannon, with cannon and ammunition conditioned at the temperature specified in Table III,

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for each five-round group, Ammunition shall be inert loaded M106 projectile with dummy M73 fuze, M82 primer, and Zone 9 M188a1 propelling charge. Bore - scope (see 4.8.1.3.1), and magnetic particle (see 4.8.1.3.2) examinations shall be conducted on each cannon after firing in accordance with the firing schedule of Table 111. A Muzzle brake is required when using Zone 9 ammunition.

TABLE 111. Climatic test firing schedule.

Sequence	No, of rounds	Temperature of Carmen & ammunition
1	5	+125%
2	5	- 25°F
3	5	- 500F

4.8.3.4 Reliability test. The reliability test shall be conducted on two cannon. Each cannon shall be fired 1288 times for a total of 2576 EFC rounds. Ammunition shall be inert loaded M106 projectile with dummy M73 fuze, M82 primer, and Zone 9M188Al propelling charge at 70°F. A muzzle brake is required when using Zone 9 ammunition. Rounds fired in other tests of 4.8 may be counted as rounds for this test. In order to demonstrate that the reliability requirements have been achieved with 80 percent confidence, this test shall be completed with no failures. Failure scoring criteria for assessing reliability requirments shall be in accordance with the APPENDIX. Each time a failure occurs during testing, the procedures outlined in 4.9 shall be completed. The Mean Rounds Between Failure (MRbF) shall be calculated using the following formula:

$$MRBF = \frac{\text{Total EFC Rounds Fired}}{\text{Number of Failures}}$$

NOTE: If no failures occur, it is not necessary to calculate the MRBF.

4.8.3.5 Durability test. The durability test shall be conducted on two cannon. Ammunition shall be inert loaded M106 projectile with dummy M73 fuze, M82 primer, and Zone 9 M188A1 propelling charge at 70F. Rounds fired in the re- liability test (see 4.8.3.4) may be conducted towards the rounds necessary for this test. A muzzle brake is required when using Zone 9.

4.8.3.5.1 Tube Wear. Durability shall be demonstrated by measuring wear after every 250 rounds until 0.135 inch bore wear has occured at 46.0 inches from the rear face of the tube. The firing rate shall not exceed one round every two minutes. If the preceding wear occurs in either or both tubes before 3200 EFC rounds have been reached, 4. 10 applies.

4.8.3.5.2 Muzzle brake (M201A1 cannon onlv). After completion of durability tube wear test rounds on each cannon, the muzzle brake shall be magnetic particle inspected in accordance with applicable drawings and specifications. If sound- ness requirements are not met. 4.9 applies.

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4.8.3.5.3 Breech ring assembly and breechblock assembly. The breech ring and breechblock assemblies shall be tested simultaneously for durability. Durability shall be demonstrated by hydraulically pressurizing six assemblies. Simulation proof test equipment (see 4.8.3.2) shall not be used for durability tests. The assemblies shall be cycled to failure at 44800 + 1600 psi. Demonstrated durability shall be calculated for 0.1% failures assuming a two parameter lognormal distribution. The calculated durability shall satisfy the requirments of 3.2.2.2. If requirement are not met, 4.9 applies.

4.9 Failure analysis. The contractor shall assist in the failure analysis procedure by performing a dimensional, physical, and visual examination of the components which are suspected to have caused the failure. Results shall be submitted to the Government for a failure cause determination. If it is determined that failure occurred because of deviation from specified dimensions, or physical properties, or both, the contractor shall accomplish, as applicable, the following action:

a. Evaluate and correct the applicable production processes and procedures to prevent recurrance of the same defect in future production.

b. Examine comrletely assembled cannon and partially assembled cannon and components to ensure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

c. Submit the corrective actions taken to the Government for evaluation and approval of retest or continuation of test.

4.10 Tube wear data. If tube wear requirements are not met (see 3.2. Z. 2) , the results shall be analyzed in accordance with currently available wear data for the M201/M20lA1 tube and a determination made as to whether the results indicate performance substantially below expected levels. If it is determined the results so indicate, then the following procedures shall apply:

a. Conduct a study performing a dimensional, visual and mttalwgical examination of the tube.

b. Review the ammmunition elememts (ProPellant) charge, and projectile characteristics) for abnormalities or changes in design or confirmation since last tested.

c. Evaluate applicable production processes and procedures and correct as warranted.

d. Examine tubes at in-process and final assembly point tO assure that defects are purged from the inventory and are not presented to the Government for acceptance.

e. Submit results of the study and corrective actions taken to the Government for evaluation and approval to retest or continue test.

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5. Packaging

5.1 First article. When required in the contract, a first: article package shall be submitted in accordance with 3.1. The item shall be unit packaged for the level of protection specified in the contract and packed Level B in accordance with the applicable packaging data sheet or MIL-P-14232. Markig shall be as specified on the packaging data sheet.

5.2 Cannon. The cannon shall be packaged, packed and marked in accordance with the requirements of packaging data sheet P11578719 for the 8 inch How. M201 and P11579521 for the 8 inch How. X201A1 for the level of protection specified in the contract (see 6.1),

5.3 Repair parts, tools and equipment. The items shall be unit packaged, packed and marked in accordance with the applicable packaging data sheet and MIL-P-14232 for the level of protection specified in the contract (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurment requirements.

a. Title, number and data of this specification.

b. Requirements for submission of the shipping instructions for first article (see 3.1 and 4.4) and first article package (see 3.1 and 5.1).

(1) The contract should designate the examinations and tests *to* be performed by the contractor and the examinations and tests to be performed by the Government.

(2) When different that that specified in para. 4.4.1, the contract should specify the sample size (i.e. , the number of units) of the first article.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(4) The contract should designate at whose expense a retest may be performed.

(5) The contract should specify that the acceptable first article shall be delivered in accordance with the terms of the contract.

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(6) Any additional or extended examinations and tests beyond the scope of this specification should be specified and identified as a separate item of the contract. Tests may be conducted on cannon or critical assemblies or parts thereof when: (1) anew contractor is utilized, (2) a time lapse has occurred since the item has last produced by the contractor, (3) significant changes have been made in a manufacturing process Or material, Or (4) a design change is made. The determination as to the significance of the action with regard to the requirement for testing shall be made by the product assurance element of the procuring activity. Critical items are: (a) tube, (b) breech ring, (c) breech bushing, (d) breechblock, (e) muzzle brake (M201Al).

c. The serial numbers for cannon tube, firing mechanism, and muzzle brake (for M201A1 only).

d. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.7.3).

e Availability of Government inspection equipment.

f, Extent of supplier's responsibility for: Government-tiished and supplier-required final inspection equipment.

g Applicable acceptance test procedures, initial consecutive quantity (i) and sampling frequency (f) (see 4. 7.6).

h. The contract should specify under what conditions simulated proof acceptance May be performed, the test equipment available, and the procedures to be followed.

i The levels of preservation, packaging, and packing required (see 5). Packaging of cannon for level B protection during domestic shipments and covered storage, shall be limited to immediate use which is not to exceed 90 days from date of processing.

j Packaging instructions for repair parts (see 5.3) .

6.1.2 Contract data requirements, Monthly reports of the results of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

6.2 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-1-45208, as appropriate, on the Management Control Systems Summary List DD-Form 1660.

6.3 Unless otherwise specified, the contract should specify the application of MIL-1-45607 and MIL-STD-45662 on the **Management** Control Systems Summary List DD Form 1660.

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APPENDIX

SCORING CRITERIA FOR ASSESSING RELIABILITY REQUIREMENTS OF CANNON, 8 INCH HOWITZER: M201/m201Al

10. SCOPE

10.1 Scope. This appendix provides Reliability assessors guidelines for the consistent classification of failure for assessing the Reliability Requirements of the Cannon 8 inch Howitzer: M201/M201Al as stAted in the "Development Plan for m110E2 inch Improved SPH" dated Nov. 1975.

20. APPLICABLE cable documents

PUBLICATIONS

developmnt Plan for M110E2 8 inch Improved SPH dated Nov. 1975
m110a2- scoring Criteria dated 5 Aug 1975

30. SCOrING CRITERIA

30.1 Basis. This scoring criteria is based on. the reliability scoring procedure approved by the Department of the Amy. It was extracted from the M110A2 Scoring Criteria" dated 5 Aug 1975. The Failure Scoring Procedure (see 30.2) provides for the use of the Failure definition (see 3.2.1.3) which is amplified (see 30.1.1), Failure Criteria (see **30.1.2),** and Degraded Modes of Operation (see 30.1.3) .

30.1.1 Failure definition amplification.

30.1. 1.1 The following are not considered as reliability failures:

1. Scheduled replacement of parts before failure.

2. An incipient malfunction corrected during scheduled preventive maintenance on the part in question provided a higher level of maintenance is not necessary.

3. A malfunction resulting from not following the prescribed operational or maintanance procedures or schedule dictated by the equipment manuals.

4. malfunction resulting from test item abuse, unrealistic operating conditions or accident.

5. Actual or incipient malfunctions detected or corrected during initial technical inspections and incipient malfunction detected during final technical inspection.

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30.1.1.2 The following are considered as reliability failures:

1. A failure detected and corrected during the correction of another failure provided the failures are totally unrelated.

2. Corrected incipient malfunctions not covered by 30.1.1 1 **examples** 2 and 5.

3. Failures detected and connected during final technical inspection.

4. Failures resulting from lack of clarity of instruction or other fault in the maintenance test package.

30.1,2 Failure males. The following is a list of typical failure modes for use in assessment of mission reliability failures. These failure males **are** subject to performance de-gradation (see 30.1.3).

1. Breechblock cannot be closed or opined.
2. Gun will not fire.
3. Primer cannot be removed or inserted.

30.1.3 degraded males of operation. A certain amount of performance degradation must take place before a reliability failure can be assumed. Of course,-if the system ceases to function, performance becomes unacceptable (i.e. , non-existent); however, it performance is only degraded, then some threshold must be established to define where reliability failures begin. Allowable degradation thresholds have been established for the various performance parameters. Performance beyond these thresholds constitutes a reliability failure. As long as performance remains above the threshold or is returned above the threshold within five minutes maintenance period, no reliability failure is charged.

1. Rate of Fire -	4.5 rounds/3min
2. Rate of Fire - sustained	1 round/2ti

30.2 Failure scoring procedure. The following scheme provides the instructions for the use of the Failure Definition (see 3.2. 1.3) for assessing the reliability requirements of the cannon, 8-inch Howitzer: M201Al (see 3.2.2.1). These instructions assume that an incident or malfunction has occured during test, and a record has been prepared against this action to the extent where it is necessary to determine the chargeability of the action. Incidents where the identification of a reliability failure is in doubt will be deferred to a user-developer-tester Scoring Conference. The step-by-step instructions for using the Failure Definition are as follows:

Step 1. Record the incident information when malfunction of the cannon occurs or is detected (rounds on test item and system, description of incident, nature of maintenance action required, maintenance

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time in 'hours, maintenance manhours, etc). Proceed to the next step answering the question posed and taking action according to the answer given.

Step 2. Is this a scheduled replacement of parts before fail ure? If yes, do not charge a failure; if no, proceed to Step 3.

Step 3. Is this a malfunction resulting from not following the prescribed operational or maintenance procedures or schedule-dictated by the equipment manuals? If yes, do not charge a failure; if no, proceed to Step 4.

Step 4. Is this a malfunction resulting from test abuse, unrealistic operrating conditions or accident? If yes, do not charge a failure; if no, proceed to Step 5.

Step 5. Is this an actual or incipient malfunction detected or corrected during initial technical inspection or an incipient malfunction detected during final technical inspection? If yes, do not charge a failure; if no, proceed to Step 6.

Step 6. Is this an incipient malfunction corrected during scheduled preventative maintenance on the part in question, provided a higher level of maintenance is not necessary? If yes, do not charge a failure; if no, proceed to Step 7.

Step 7. Is this incident related to or caused by another reliability failure? If yes, do not charge a failure, if no, proceed to Step 8.

Step 8. The following statements are addressed collectively (A) The malfunction did not and would not cause a critical or catastrophic hazard to personnel or equipment as defined by MIL-STD-882. (B) The system performance is above acceptable levels (see 30.1.3). (C) The malfunction did not and would not preclude the ability to comence or the cessation of any mode of operation (e.g. , load and fire at any range, charge, elevation, or direction), (see 30.1.2). If all three of the above statements are true, do not charge a reliability failure; if any are not true, proceed to Step 9.

Step 9. Is the operator, crew authorized and able to remedy the malfunction by adjustement repair, or replacement action within five minutes, using the controls, Basic Issue Items (BII), Items Troop Installed or Authorized (ITIA) and parts authorized to the crew? If the answer is yes, do not charge a reliability failure.

If the previous 9 steps provide inconclusive evidence as to the chargebility or non-chargeability of a reliability failure, the incident should be checked against the criteria of 30.1. If categorization of the incident remains in doubt, defer classification to a user-developer-tester Scoring Conference.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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MIL-C-63541 (AR)
NOTICE 1
3 August 1987

MILITARY SPECIFICATION

CANNON 8 INCH HOWITZER: M201/M201A1

This notice should be attached to MIL-C-63541 (AR) dated 22 January 1982.

MIL-C-63541 (AR) has been reviewed and determined to be current as of 3 August 1987 and may be used for acquisition purposes.

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